VU Medical Zone

CHE 301 Current Final Term Spring 2021

Q1. Give the base value of Carboxylic acid and ester? 2 Marks

The base value of Carboxylic acid and ester is **195.**

Q2. Why molecule display color? 2 Marks

Transition metals are coloured because of d-d transitions while organic molecules are coloured because of conjugation. Higher wavelength or lower energy.

Q3. What are the criteria for a molecule to be absorbed in UV range? 3 Marks

For an absorption in normal UV range the molecule must have conjugated system or non-bonding electron along with double bond.

Q4. Write some parts of UV spectrophotometer? 3 Marks

UV Spectrophotometer:

• Light source

• Monochromator

• Sample

compartment

• Detector

• Read out device

Q5. Non-polar interaction of solid phase extraction? 3 Marks

• Also called hydrophobic or reverse phase

• Involves van der Waals / dispersion forces

• Sorbents - C2, C3, C4, iC4, tC4, C5, C6, C7, C8, C10, C12, C18, C20, C30 phenyl and cyclohexyl

• Applications - drugs of abuse, pesticides

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• Analytes - protonated / neutral state, aromatics & alkyl chains

• Matrix - biologicals, water, aqueous buffers

• Elution solvents - typically non-polar to moderately polar

Q6. Applications of IR Spectroscopy. 5 Marks

Applications of IR Spectroscopy

1. Structure Determination

• Search for benzene ring presence

• Search for functional group regions (carbonyl, NH, OH, CO etc.)

• Search for CH of different types

2. Identification of compound

• Matching given structure (metal complex formation)

• Matching of standard through finger print region

• Differentiate if we have alcohol or ether

3. Detection of impurities

• Cyclohexanone is readily detectable in cyclohexanol

• Based on functional groups or finger print region

Q7. Complementary Color? 3 Marks

When white light falls upon a sample, either reflection or absorption takes place. If only a portion of the light is absorbed, the color of the sample is determined by the reflected light (is called complementary colours).

Q8. Working of ICP? 3 Marks

Working of ICP:

• The sample is prepared in liquid form as in case of FAAS.

• Sample is injected through nebulizer @ 0.5 to 1.5 mL/min.

• The emitted spectral lines pass through monochromator.

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• The separation in MS is on basis of mass to charge ratio.

Q9. Write a note on TLC. 5 Marks

Thin layer Chromatography

• Stationary Phase used: adsorbent silica pasted on glass or aluminum plates

• Mobile Phase used: usually organic solvents (pure or mixed in variable ratios)

• TLC plates are made of glass or aluminum

• Solvent equilibrates the environment

• May be placed with wall so that it may not fall in the chamber

• Separations in TLC involve distributing a mixture of two or more substances between a stationary phase and a mobile phase.

• The stationary phase is a thin layer of adsorbent (usually silica gel or alumina) coated on a plate.

• The mobile phase is a developing liquid which travels up the stationary phase, carrying the samples with it.

• Components of the samples will separate on the stationary phase according to "how much they adsorb on the stationary phase versus how much they dissolve in the mobile phase" (affinity).

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